

Trigon

A Modern, elegant and versatile theme for Beamer

Thomas Lambert
trigon@thl.ovh

v0.7.0 — 2023/02/20

Contents

1	Introduction	2
2	Getting Started	2
2.1	Installing from CTAN	2
2.2	Installing from Gitlab	2
2.3	Overleaf	3
2.4	A Minimal Example	3
2.5	Dependencies	4
3	Customization	4
3.1	Package options	4
3.2	Color Customization	7
3.3	Logos	7
4	Tips & Tricks	8
4.1	Backup Slides	8
5	Known Issues	8
5.1	Title formats	8
6	License	9
7	Implementation	9
7.1	Trigon parent theme	9
7.2	Trigon inner theme	11
7.3	Trigon outer theme	24

7.4	Trigon font theme	27
7.5	Trigon color theme	33

1 Introduction

Trigon found its origin and inspiration in the graphical guidelines resulting from the visual identity overhaul of the [University of Liège](#) (Belgium).

Although directly inspired from these guidelines, Trigon was stripped out of any mention or specificities related the University and its faculties. This makes the Trigon theme perfectly suitable for many different contexts.

The final product provides a modern, elegant and versatile theme with a high degree of customization.

Trigon’s codebase lives on [Gitlab](#). The implementation of this theme is strongly inspired from the excellent [Metropolis](#) theme by Matthias Vogelgesang, from which it borrows numerous options.

Any feature request, issue report or merge requests are welcome.

2 Getting Started

2.1 Installing from CTAN

The latest stable version of Trigon is available on [CTAN](#) and should now be part of the usual $\text{\TeX}$ distributions ($\text{\TeX}$ Live, $\text{\MacTeX}$, $\text{\MikTeX}$), under the name `beamertheme-trigon`. It means that if your distribution is kept up-to-date, the package should normally be already installed on your system. If this is not the case, consider updating the packages of your $\text{\TeX}$ distribution.

For $\text{\TeX}$ Live and $\text{\MacTeX}$ users, this usually means running

```
tlmgr update --all
```

, or if administrative privileges are required

```
sudo tlmgr update --all
```

For $\text{\MikTeX}$ users, please refer to [the official \$\text{\MikTeX}\$ documentation](#).

2.2 Installing from Gitlab

If you want to use the cutting-edge development version of Trigon, you can install it manually by following these steps:

Download the source from [Trigon repository](#) using `git clone` or as a [zip archive](#) of the latest development version.

Compile the style files by running `make sty` inside the downloaded directory. (Or run L^AT_EX directly on `source/trigontheme.ins.`)

Move the resulting `*.sty` files to the folder containing your presentation. To use Trigon with many presentations, run `make install` or move the `*.sty` files to a folder in your T_EX path instead.

Use the theme by declaring `\usetheme{trigon}` in the preamble of your document.

Trigon uses the Make build system to offer the following installation options for advanced users:

`make sty` builds the theme style files.

`make doc` builds this documentation manual.

`make demo` builds a demo presentation to test the features of Trigon.

`make all` builds the theme and manual.

`make clean` removes the files generated by `make all`.

`make install` installs the theme into your local texmf folder.

`make uninstall` removes the theme from your local texmf folder.

2.3 Overleaf

Since October 2021, [TexLive 2011 is available on Overleaf](#). Trigon is therefore automatically supported for new documents. If you want to switch older documents to use the Trigon theme, you need to [select TexLive 2021 or higher as your TeX Live version](#) (this may cause issues with other packages used in your presentation).

In addition to that, a template project was created and is publicly available in the [Overleaf Gallery](#) to bootstrap your presentation.

2.4 A Minimal Example

The following code shows a minimal example of a Beamer presentation using Trigon.

```
\documentclass{beamer}
\usetheme{trigon}      % Use trigon theme
\title{A minimal example}
```

```

\subtitle{A subtitle example}
\date{\today}
\author{Thomas Lambert}
\institute{Your university or company}
\begin{document}
  \maketitle
  \section{First Section}
  \begin{frame}{First Frame}
    Hello, world!
  \end{frame}
\end{document}

```

2.5 Dependencies

Trigon depends on the `beamer` class and the following standard packages:

- `tikz`
- `pgfopts`

The theme works best with the open source [Source Sans Pro](#) font from Adobe. Therefore, the package `sourcesanspro` is loaded by default with the theme. However, if the user prefers to select an other font, the theme option `nosourcefonts` can be used to prevent the font package from being loaded.

3 Customization

3.1 Package options

Trigon provides a number of options, which can be set using a key=value interface. The primary way to set options is to provide a comma-separated list of option-value pairs when loading Trigon in the preamble:

```
\usetheme[option1=value1, option2=value2, ...]{trigon}
```

Options can be changed at any time — even mid-presentation! — with the `\trigonset` macro.

```
\trigonset{option1=newvalue1, option2=newvalue2, ...}
```

The list of options is structured as shown in the following example.

option key [list of possible values](#) default

A short description of the option.

As Trigon implementation is heavily inspired from the excellent [Metropolis](#) theme by Matthias Vogelgesang, many of Metropolis options are also available in Trigon.

3.1.1 Main theme

background [light,dark](#) light

Provides the option to have a dark background and light foreground instead of the reverse.

titleformat [regular, smallcaps, allsmallcaps, allcaps](#) regular

Changes the format of titles, subtitles, section titles, frame titles, and the text on “standout” frames. The available options produce Regular, SmallCaps, allsmallcaps, or ALLCAPS titles. Please refer to Section [5.1](#) for known issues with these options.

usesourcefonts [true,false](#) true

Chooses if the `sourcesanspro` font package should be loaded `true` or not `false`.

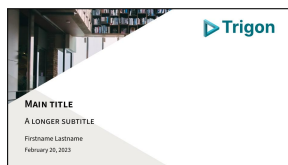
3.1.2 Inner theme

sectionpage [none, simple](#) simple

Adds a slide at the start of each section (`simple`). The `none` option disables the section page.

titlestyle [plain, style1, style2](#) style2

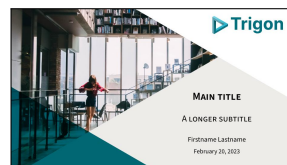
Changes the layout of the title page.



plain



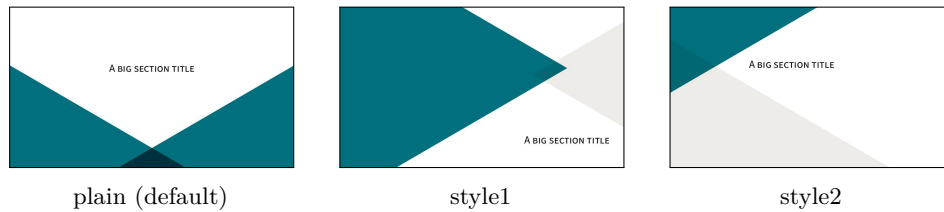
style1



style2 (default)

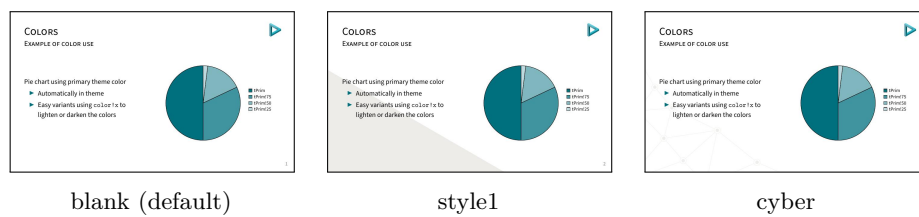
sectionstyle [plain, style1, style2, style3](#) plain

Changes the layout of the section page. `style3` is similar to `plain` but with the right triangle in grey instead of theme color.



slidestyle `blank, style1, cyber` `blank`

Changes the background of the regular frames.



3.1.3 Outer theme

numbering `none, counter, fraction` `counter`

Controls whether the frame number at the bottom right of each slide is omitted (`none`), shown (`counter`) or displayed as a fraction of the total number of frames (`fraction`).

3.1.4 Color theme

block `transparent, fill` `fill`

Optionally adds a light grey background to block environments like `theorem` and `example`.

headingcolor `default, theme` `default`

Controls whether the color of all headings (title page, frame title, etc.) should be in black (`default`) or in a slightly darker shade of the theme color `theme`.

textcolor `default, theme` `default`

Controls whether the color of the regular text elements should be in black (`default`) or in a slightly darker shade of the theme color `theme`.

3.1.5 Font theme

<code>titleformat title</code>	<code>regular, smallcaps, allsmallcaps, allcaps</code> <code>regular</code>
<code>titleformat subtitle</code>	Individually controls the format of titles, subtitles, section titles, and frame titles
<code>titleformat section</code>	(see <code>titleformat</code> , above).
<code>titleformat frame</code>	

3.2 Color Customization

The included Trigon color theme is used by default, but its colors can be easily changed to suit your tastes. All of the theme's styles are defined in terms of a few main colors:

- `tPrim` The primary theme color, used for main triangle elements, and for headings and text if the appropriate options are selected.
- `tSec` The secondary theme color (lighter shade of the primary color), used for some triangle elements and example text.
- `tAccent` The accent color, used mostly for alerted text.
- `tGrey` A grey color, used for background elements (triangles, blocks, etc.).
- `tTxt` The default text color.
- `tBg` The default background color.
- `tGreyBg` A tint of grey to use for the triangle elements.
- `tTheme` A tint of the `tPrim` color to use for some triangle elements.

An easy way to customize the theme is to redefine these colors using

```
\colorlet{tPrim}{ ... }
\colorlet{tSec}{ ... }
...
```

in your preamble.

3.3 Logos

The package allows the inclusion of logos on the title page and on the content slides. For that, the following commands are defined:

```
\biglogo[<options>]{path/to/your/titlepage_logo}
\smalllogo[<options>]{path/to/your/titlepage_logo}
```

These two commands are just replacements for `\includegraphics` and they will pass the options as-is so you can perfectly size and position the images.

4 Tips & Tricks

4.1 Backup Slides

Speakers will often include extra slides at the end of their presentation to refer to during audience questions. One easy way to do this is to include the `appendixnumberbeamer` package in your preamble and call `\appendix` before your backup slides.

Trigon will automatically turn off slide numbering for slides in the appendix.

5 Known Issues

5.1 Title formats

Be aware that not every font supports small caps, so the `smallcaps` or `allsmallcaps` options may not work if you use a font other than `sourcesanspro`.

The title format options `allsmallcaps` and `allcaps` are quite nice from an aesthetic point of view, but their use of `\MakeLowercase` and `\MakeUppercase` can cause unexpected problems. For example:

- Some commands, like `\`, do not work inside `\MakeLowercase` and `\MakeUppercase`. (See Metropolis [#125](#))
- Only alphabetic characters are affected by `\MakeLowercase`, so numerals and punctuation remain at full height. This can spoil some of the aesthetic benefits of `allsmallcaps`. (See Metropolis [#33](#))
- `\MakeLowercase` and `\MakeUppercase` apply to math mode and `\scshape` does not. This can easily introduce mathematical errors that are hard to catch.
- It is impossible to typeset symbols which are encoded as uppercase letters in a different font. In particular, `\mathbb` and `\mathcal` letters will be replaced by other math glyphs. (See Metropolis [#153](#))

The `allsmallcaps` and `allcaps` options are safe to use if your titles contain only alphabetic characters and do not require the expansion of any macros.